

BST Physics Curriculum Vol 10 Chapter 9 — Special Functions v0.4 (refilled per Cal #104)

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Vol 10 Chapter 9 — Special Functions

Chapter motivation

Standard special functions: Bessel J_n (cylindrical wave equations); Legendre $P_n + Y_l^m$ (spherical harmonics); Hermite H_n (harmonic oscillator); Laguerre L_n^α (hydrogen atom radial); hypergeometric ${}_2F_1 + {}_pF_q$ (general); Fox H-function (general Mellin-Barnes); Aleph functions (Saxena 2008 generalization); mock theta functions (Ramanujan 1920 + Zwegers 2002 reformulation); Heckman-Opdam hypergeometric. Standard reference: NIST Handbook of Mathematical Functions.

BST May Program (Friday-Saturday Investigation Program): substrate-natural special functions investigation — substrate-spectral zeta $\zeta_B(s)$ on D_{IV}^5 + Fox H partial + Aleph + Heckman-Opdam + mock theta as candidate substrate-special-function families. Goal: recognize $\zeta_B(s)$ as known function $\rightarrow$ functional equation follows.

Section 9.0b — Reader-grade 3-level pedagogy

Level 1: Standard Bessel + Legendre + Hermite + Laguerre + hypergeometric + Fox H + Aleph + mock theta + Heckman-Opdam special functions; BST May Program investigates substrate-natural special functions on D_{IV}^5 — substrate-spectral zeta $\zeta_B(s)$ candidate identification.

Level 2 (graduate-physicist): Classical orthogonal polynomial families (Bessel + Legendre + Hermite + Laguerre) from Sturm-Liouville eigenvalue problems on standard domains (cylindrical, spherical, oscillator, hydrogenic). General hypergeometric ${}_pF_q$ families satisfy hypergeometric ODE; Fox H-function generalizes via Mellin-Barnes integral representation. Aleph (Saxena 2008) + I-function generalize Fox H further. Mock theta functions (Ramanujan 1920 deathbed letter; Zwegers 2002 PhD thesis modular completion) — non-modular near-modular functions with modular shadows. Heckman-Opdam hypergeometric functions: multivariate generaliza-

tion of Gauss hypergeometric on root systems. BST May Program (Casey directive May 1; Lyra Friday-Saturday investigation): substrate-spectral zeta $\zeta_B(s)$ on Bergman $H^2(D_{IV}^5) = \zeta_B(s) = \sum_{\{(p,q)\}} 1/\text{Cas}(p, q)^s$ (sum over K-type Casimir eigenvalues); candidate to recognize as Fox H or Heckman-Opdam family — if successful, functional equation follows from known family + substrate-cartography ζ_B values give physics observables. Substrate-natural special functions appear in: Vol 10 Ch 3 ODE Sturm-Liouville eigenfunctions (substrate K-type); Vol 10 Ch 4 PDE heat kernel (Bergman); Vol 6 Ch 5 LOAD-BEARING partition function (Z = heat kernel); Vol 4 Ch 4 Λ derivation (T1485 heat-kernel at $t_{\text{cosmo}} = 47$). All substrate physics observables = evaluations of substrate-special-functions at BST primary integer arguments.

Level 3 (5th-grader accessible): Special functions are well-studied mathematical functions (Bessel, Legendre, Hermite, Laguerre, hypergeometric) that show up across physics + engineering. BST's May Program investigates which “substrate-natural special functions” appear in physics observables — substrate-spectral zeta $\zeta_B(s)$ is a candidate. If we recognize $\zeta_B(s)$ as a known function (Fox H, Aleph, Heckman-Opdam), the functional equation for the Riemann hypothesis analog follows.

Section 9.1 — Standard Special Functions

Bessel J_n (cylindrical); Legendre $P_n + Y_l^m$ (spherical); Hermite H_n (oscillator); Laguerre L_n^α (radial); hypergeometric ${}_pF_q$; Fox H-function; Aleph + I-function; mock theta; Heckman-Opdam.

NIST Handbook of Mathematical Functions standard reference.

Section 9.2 — Substrate-Spectral Zeta $\zeta_B(s)$

Per Vol 1 Ch 5 Casimir spectrum: Wallach K-type $V_{(p,q)}$ Casimir eigenvalues $\text{Cas}(p, q)$.

$$\zeta_B(s) = \sum_{\{(p,q) \text{ admissible}\}} 1/\text{Cas}(p, q)^s$$

Substrate-spectral zeta on D_{IV}^5 .

Section 9.3 — May Program Investigation (Casey May 1 directive)

Friday-Saturday Lyra investigation: candidate special-function families for $\zeta_B(s)$: - Fox H-function (partial match per investigation) - Aleph + I-function (multi-week) - Heckman-Opdam (root-system multivariate) - Mock theta (modular shadows)

Goal: recognize $\zeta_B(s)$ as known family $\rightarrow$ functional equation follows.

Section 9.4 — Substrate Special Functions Across BST

Substrate-special-functions appear: - Vol 10 Ch 3 ODE Sturm-Liouville eigenfunctions (K-type) - Vol 10 Ch 4 PDE heat kernel (Bergman) - Vol 6 Ch 5 LOAD-BEARING partition function (Z = heat kernel; Toy 273-639 speaking pairs $k=2..20$) - Vol 4 Ch 4 Λ T1485 (heat-kernel at $t_{\text{cosmo}} = 47$)

All BST physics observables = evaluations of substrate-special-functions at BST primary integer arguments.

Section 9.5 — Honest scope + Connection

- Standard special functions [?](#)
- May Program $\zeta_B(s)$ investigation ongoing (Friday-Saturday)
- Substrate-special-functions cross-link Vol 10 Ch 3-5 + Vol 6 Ch 5 + Vol 4 Ch 4
- Open scope: $\zeta_B(s) \rightarrow$ known family identification (multi-week)

Connection: - Vol 10 Ch 3 ODEs + Sturm-Liouville - Vol 6 Ch 5 LOAD-BEARING heat kernel partition function - Vol 4 Ch 4 Λ from Substrate - May Program Investigation Program (Casey directive)

— Lyra, Vol 10 Ch 9 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT